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Graph-Based Multi-Region Framework for MRI-Based Classification of Progressive Mild Cognitive Impairment

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Graph-Based Multi-Region Framework for MRI-Based Classification of Progressive Mild Cognitive Impairment

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Abstract

Distinguishing progressive mild cognitive impairment (pMCI) from stable mild cognitive impairment (sMCI) using structural MRI remains difficult because disease-related anatomical changes are subtle, heterogeneous, and spatially distributed. We propose a graph-based multi-region framework that represents deformation-derived MRI biomarkers as anatomically organized graph signals for sMCI versus pMCI classification. Structural information was characterized using displacement magnitude, Jacobian determinant statistics, and strain measures derived from the Frobenius norm of the infinitesimal strain tensor. Descriptors were extracted from fifteen brain regions, with hemispheres modeled independently. A fixed population-level graph template was estimated from stable inter-regional correlation patterns without class labels, predictions, or classification outcomes. Subject-specific regional descriptors were then used as node attributes in a Graph Attention Network, integrating deformation information across anatomically related regions. The framework was evaluated on 469 MCI subjects from ADNI, aged 60–90 years. Using the 60% edge-frequency template, the method achieved AUC/accuracy/F1-score of **0.92 ± 0.03/81.21 ± 4.47%/0.79 ± 0.04** for males and **0.88 ± 0.03/78.41 ± 4.39%/0.73 ± 0.07** for females, outperforming conventional baselines trained on the same features. These findings indicate that deformation-informed anatomical graphs provide an effective representation for MRI-based pMCI classification.

Keywords: Alzheimer’s disease, mild cognitive impairment, graph neural networks, deformation-based morphometry, magnetic resonance imaging.

1 Introduction

Alzheimer’s disease (AD) is a progressive neurodegenerative disorder and a leading cause of dementia worldwide. Because neuropathological changes may precede clinical diagnosis by several years, identifying subjects with Mild Cognitive Impairment (MCI) who are more likely to progress to AD is an important research problem in neuroimaging. Distinguishing stable MCI (sMCI) from progressive MCI (pMCI) is particularly challenging because structural alterations in individual subjects are often subtle, heterogeneous, and distributed across multiple brain regions rather than confined to a single anatomical site.

Large-scale initiatives such as the Alzheimer’s Disease Neuroimaging Initiative (ADNI) have supported the development and benchmarking of imaging biomarkers and predictive models for AD progression [35]. Traditional magnetic resonance imaging (MRI)-based approaches commonly rely on voxel-based morphometry (VBM) and related computational anatomy techniques to quantify local or regional atrophy patterns [1], as well as surface-based morphometry (SBM) pipelines to characterize cortical alterations using tools such as FreeSurfer [8, 10]. These methods have provided important evidence about structural patterns associated with AD progression. However, in many predictive pipelines, regional measurements are ultimately treated as independent or weakly structured feature vectors, which may limit the ability to represent coordinated anatomical alterations across distributed brain systems.

Deformation-based morphometry provides a complementary strategy to characterize anatomical differences through displacement vector fields, Jacobian determinants, and derived measures of local deformation. Such descriptors summarize spatial displacement, local volumetric variation, and deformation intensity in anatomically defined regions, and have been shown to carry predictive value in MRI-based studies of conversion from MCI to AD [23]. Once these regional descriptors are extracted, a central methodological question is how to organize them so that both local morphometric information and inter-regional relationships can contribute to classification.

Graph-based learning offers a natural representation for this problem by treating anatomical regions as nodes and their relationships as edges. Graph Neural Networks (GNNs), including Graph Convolutional Networks (GCNs) [16] and Graph Attention Networks (GATs) [33], enable message passing across connected regions and can capture network-level patterns that are not explicitly represented by independent regional features. In neuroimaging, population-graph strategies have shown that combining imaging features with graph-based modeling can improve disease prediction performance [26]. These models are therefore well suited to integrate deformation-derived regional biomarkers within an anatomically structured representation.

In this work, we propose a graph-based multi-region framework for MRI-based classification of pMCI and sMCI that integrates deformation-derived regional descriptors with fixed population-level graph templates. The templates are estimated through

a label-free graph stability analysis based on inter-regional correlations, and classification is performed using subject-specific node attributes under a shared graph topology. We also evaluate sex-specific graph templates because male and female cohorts may present distinct population-level patterns of structural organization. The main contributions of this study are: (i) a deformation-based, multi-region feature design combining displacement magnitude, Jacobian determinant, and strain-derived measures; (ii) a stability-based graph construction protocol for representing recurrent inter-regional deformation relationships; (iii) a GAT-based classifier operating on subject-specific node attributes under a shared anatomical graph structure; and (iv) an empirical analysis of sex-specific graph organization patterns observed during sMCI versus pMCI classification.

2 Related Work

Early prediction of AD progression from MCI using structural MRI has been extensively investigated through machine learning, deep learning, and computational anatomy approaches. Traditional pipelines commonly rely on VBM, SBM, or hand-crafted regional features combined with classical classifiers. Moradi et al. [24] proposed a semi-supervised framework integrating MRI-derived biomarkers with demographic and cognitive variables, achieving an area under the receiver operating characteristic curve (AUC) of 0.90 for sMCI versus pMCI classification. Such studies demonstrate that structural MRI contains useful information for progression prediction, while also motivating representations that can integrate multiple regional measurements in a structured manner.

Longitudinal MRI-based approaches have demonstrated sensitivity to disease progression by explicitly modeling temporal change. Hu et al. [12] introduced a hybrid convolutional neural network (CNN)–Transformer architecture using dual time-point MRI and multi-scale attention mechanisms, achieving an AUC of 0.86 on ADNI. Jomeiri et al. [14] proposed a DenseNet-BiLSTM framework using longitudinal MRI, achieving 83.51% accuracy for sMCI versus pMCI classification. These methods highlight the value of temporal information; the present work addresses a complementary setting by focusing on deformation-derived descriptors extracted from T1-weighted MRI and organized through graph-based modeling.

Patch-based, landmark-driven, and voxel-wise approaches have also been explored for progressive MCI classification. Majlan et al. [20] proposed a multi-stream CNN trained on statistically selected anatomical landmarks, combined with transfer learning from AD and cognitively normal subjects. Xu et al. [36] combined VBM and SBM features to reveal structural alterations in frontal, temporal, and insular regions, showing that joint morphometric indices can improve discrimination between sMCI and pMCI. These studies support the relevance of regional structural descriptors, but they do not explicitly encode deformation relationships among distributed anatomical regions.

Recent multimodal and attention-based models have further improved prediction strategies. Luo et al. [19] introduced a dual-attention fusion network combining MRI

and neurocognitive data, whereas Chen et al. [7] proposed a multimodal region-of-interest (ROI)-based fusion framework integrating structural MRI and positron emission tomography (PET). Yu et al. [37] developed a transformer-based multimodal architecture for MCI conversion prediction, and Wei et al. [34] proposed a 3D self-attention network based on structural MRI. These works demonstrate the strength of attention and fusion mechanisms in AD-related prediction tasks. The present study follows this direction from an anatomical graph perspective, using a fixed graph template to structure deformation-derived MRI descriptors before classification.

Graph-based representations have emerged as an appropriate strategy to capture inter-regional dependencies in neuroimaging. Prior studies have constructed graphs using functional, structural, or population-level relationships, often combined with graph convolutional networks for AD-related classification tasks. In contrast to approaches based on unstructured vectors or isolated regional measurements, graph models provide a direct way to encode relationships among anatomically defined regions. The framework proposed here constructs fixed, population-level graph templates from stable inter-regional deformation relationships and uses them as the message-passing domain of a GAT classifier.

Overall, existing methods have demonstrated encouraging performance for sMCI versus pMCI classification, but they differ substantially in input modalities, longitudinal requirements, clinical variables, validation protocols, and model interpretability. The proposed graph-based multi-region framework is positioned as a complementary MRI-based approach that integrates deformation-derived descriptors with an explicit inter-regional graph representation. The central question addressed in this study is whether organizing regional deformation biomarkers as anatomical graph signals improves classification relative to conventional classifiers trained on the same features.

3 Methodology

This section describes the proposed deformation-informed graph learning framework, in which subject-to-template deformation maps are transformed into regional structural biomarkers and subsequently modeled as graph-based representations of inter-regional anatomical relationships.

3.1 Overview

The classification framework illustrated in Fig. 1 aims to distinguish between stable MCI and progressive MCI subjects using MRI-derived structural descriptors. The pipeline consists of MRI preprocessing, construction of population-specific reference templates, subject–template deformation analysis, ROI-wise aggregation of deformation-derived maps, graph construction, model selection, and classification. In the deformation stage, each MCI subject is registered to two demographically matched templates, one cognitively normal and one dementia reference. The resulting deformation fields are not used as dense images; instead, they are converted into three scalar maps—displacement amplitude, Jacobian-based volumetric variation, and strain

intensity—and summarized inside predefined anatomical ROIs. These regional descriptors are then organized as node attributes in a fixed graph representation, allowing the GAT to model inter-regional dependencies during sMCI versus pMCI classification.

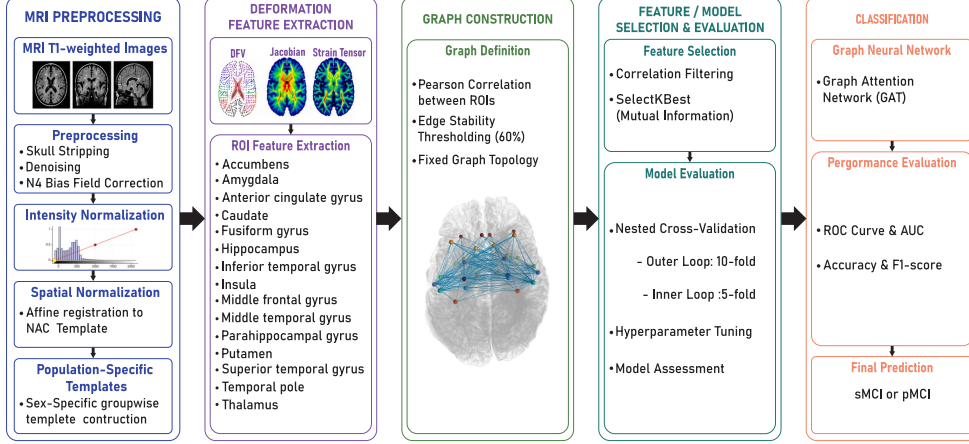


Fig. 1: Overview of the proposed framework for sMCI and pMCI classification.

3.2 Image datasets

The classification experiments in this work were conducted using MRI scans obtained from ADNI [13]. ADNI is a multicenter initiative designed to develop and validate biomarkers and imaging methodologies for the early detection and monitoring of Alzheimer’s disease using clinical, genetic, and neuroimaging data.

For this study, we selected subjects between 60 and 90 years of age diagnosed with MCI. Based on their longitudinal clinical evolution during a 36-month follow-up period, participants were divided into two groups: sMCI, comprising individuals whose cognitive condition remained stable, and pMCI, comprising subjects who converted to Alzheimer’s disease during follow-up.

The resulting cohort contains 266 subjects classified as sMCI and 203 subjects classified as pMCI. Table 1 summarizes the demographic characteristics of the participants considered in this study. Statistical comparisons were performed to assess potential differences between groups. Age was evaluated using the Mann–Whitney U test, while sex distribution was assessed using the chi-square test.

Table 1: Demographic characteristics of the cohort analyzed in this study.

	sMCI	pMCI	p-value
Number of Patients	266	203	–
Age (mean $\pm$ std)	73.4 $\pm$ 7.0	74.8 $\pm$ 7.1	0.008
Sex (F:M)	108 : 158	91 : 112	0.604

Age was not used as an explicit input variable during model training or classification; predictions were based on deformation-derived structural features extracted

from MRI data. To account for broad demographic anatomical variability during registration, population references were stratified by age and sex before subject–template deformation analysis. This design keeps the classification model focused on MRI-derived structural descriptors while reducing mismatch between subjects and reference templates used for feature extraction.

Brain aging does not follow a homogeneous or strictly linear pattern across individuals. Subjects with similar chronological ages may exhibit different structural brain characteristics, including distinct levels of atrophy, tissue deformation, and neurodegenerative progression. This variability is particularly relevant in MCI populations, where subjects with comparable ages may follow different clinical trajectories. Consequently, the proposed framework emphasizes subject-specific deformation patterns extracted from anatomically defined regions rather than using chronological age as a direct classifier input.

MRI scans from the Open Access Series of Imaging Studies (OASIS) dataset [2, 18, 21, 22] were used only for population-template construction, not for training or testing the sMCI versus pMCI classifier. The OASIS dataset includes T1-weighted MRI scans from cognitively normal subjects as well as individuals diagnosed with dementia. In this study, images from subjects aged between 60 and 90 years were selected to construct population-based templates stratified by age, sex, and diagnostic status.

For anatomical parcellation, the NAC Brain Atlas [11] was employed. This atlas was constructed from the MRI scan of a neurologically healthy 42-year-old male volunteer and developed through a collaboration between Brigham and Women’s Hospital and Massachusetts General Hospital. It provides detailed manual annotations for 359 brain structures. The T1-weighted NAC template, with isotropic resolution of 1 mm, was used as the reference anatomical space for registering the ADNI images and applying atlas-derived ROI masks.

3.3 Preprocessing

The preprocessing stage prepares structural MR images for deformation-based analysis by reducing noise, correcting intensity artifacts, standardizing intensity distributions, and aligning all scans to a common anatomical space. Each step is described below.

3.3.1 Skull stripping

Brain extraction was performed using the `brain_extraction` function from ANTsPyNet, with the modality parameter set to `t1`, as recommended for T1-weighted structural MRI. The resulting brain mask was applied to the original image through voxel-wise multiplication to generate the skull-stripped volume, which was subsequently used in all preprocessing steps.

3.3.2 Denoising

Noise suppression was performed using the Non-Local Means (NLM) algorithm implemented in ANTsPy [4]. The denoising procedure employed a patch radius of 1 voxel and a search radius of 2 voxels, with noise modeled as Rician, which is consistent with MRI magnitude data. The algorithm was applied with a shrink factor of 1,

ensuring full-resolution processing. These parameter settings are commonly adopted in structural MRI pipelines, providing an effective balance between noise reduction and preservation of fine anatomical structures.

3.3.3 Bias field correction

To correct intensity inhomogeneities caused by scanner-related artifacts, the N4 bias field correction algorithm [32] was applied using the implementation available in ANTsPy. The method estimates a smooth, low-frequency bias field through a multi-resolution optimization framework, where the image is progressively refined across scales. A B-spline representation is used to model the bias field, allowing flexible yet smooth intensity correction across the image domain. This procedure improves intensity homogeneity and reduces artifacts that may affect subsequent feature extraction.

3.3.4 Intensity normalization

Histogram matching [25] was employed to standardize intensity distributions across all subjects. The method was implemented using 256 histogram bins, following the original formulation, with the NAC template used as the reference image. This procedure reduces inter-subject variability and facilitates the extraction of comparable features.

3.3.5 Spatial normalization

All images were resampled to a uniform resolution of $256 \times 256 \times 256$ voxels and aligned to the NAC template using affine registration implemented in Elastix [17]. The affine coregistration step used the `Par0010affine` parameter file from the elastix Model Zoo, which provides a multi-resolution affine transformation optimized using Mattes mutual information. This setup was used to place all scans in a common anatomical space before ROI-based feature extraction.

3.3.6 Region definition

Anatomical ROIs were defined from the NAC atlas labels based on the grouping proposed by Planche et al. [29]. A total of fifteen regions were selected to capture anatomically and functionally relevant structures associated with neurodegenerative processes. The selected ROIs are anatomically localized in Fig. 2.

For each anatomical region, left and right hemispheres were treated independently throughout the analysis, resulting in 30 graph nodes per subject. Binary masks were derived from 50 NAC atlas labels and subsequently aggregated into the corresponding ROIs. The complete mapping between NAC atlas labels and the aggregated ROIs is provided in the supplementary material. These regions were subsequently used to extract region-wise features for all analyses in the proposed framework.

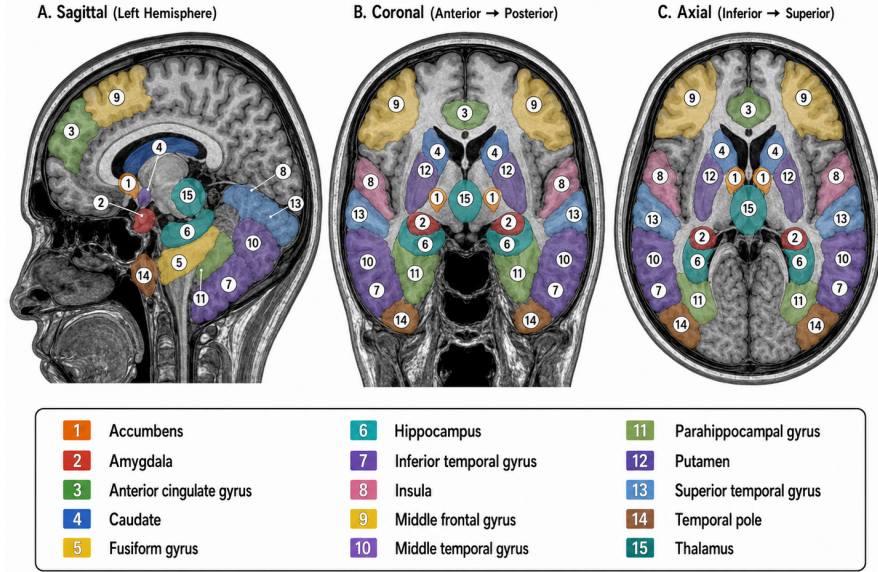


Fig. 2: Anatomical localization of the ROIs assessed in this study. The T1-weighted anatomical background is based on the NAC Brain Atlas [11]. The overlaid colors and numerical labels are schematic annotations used only to illustrate the anatomical regions assessed; they do not correspond to the exact ROI masks from the NAC Brain Atlas used for feature extraction.

3.4 Population references and subject–template deformation analysis

Population references were constructed to provide anatomically consistent targets for deformation-based characterization of MCI subjects. In contrast to a normative brain-age setting, the templates were not used to estimate chronological age. Instead, they defined two reference conditions within the same sex and age stratum: a cognitively normal reference and a dementia reference. This design allowed each MCI subject to be represented by deformation descriptors measured relative to both a non-demented and a pathological anatomical prior.

Groupwise registration was applied to MR images from the OASIS dataset [2, 18, 21, 22]. Subjects were stratified by sex, diagnostic status, and five-year age intervals between 60 and 90 years. The adopted intervals were 60–64, 65–69, 70–74, 75–79, 80–84, and 85–90 years, with subjects aged exactly 90 assigned to the last interval. Due to the reduced number of available subjects in the extreme age strata, 10 images were used for template generation in the 60–64 and 85–90 intervals, while 15 images were used for the intermediate intervals.

For each stratum, groupwise registration was performed using Elastix [17] with an adapted version of the Par0039 groupwise parameter file from the elastix Model Zoo. The original configuration was modified for the present study by

using the deformable `BSplineStackTransform`, allowing local anatomical variability among images to be modeled during template construction. The number of resolutions was set to three levels (`NumberOfResolutions` 3), providing a gradual multi-resolution optimization strategy. To improve the stability of parameter estimation, the sampling configuration was also set to `NumberOfSpatialSamples` 5000 and `NumberOfSamplesForExactGradient` 100000. Thus, the parameter file used for groupwise registration corresponds to an adapted version of the elastix Model Zoo groupwise model, configured to improve the robustness of the template-generation procedure. The resulting template represents a population-specific anatomical reference rather than an individual subject. In total, 24 templates were generated: six male cognitively normal templates, six male dementia templates, six female cognitively normal templates, and six female dementia templates. Once generated, these templates were kept fixed throughout the subsequent experiments.

Let I_s denote the preprocessed T1-weighted image of subject s . For each subject, two templates were selected according to the subject’s sex and five-year age interval: $T_{g,a}^{\text{CN}}$ and $T_{g,a}^{\text{DEM}}$, where g denotes sex, a denotes the age interval, CN denotes the cognitively normal reference condition, and DEM denotes the dementia reference condition. The subject image was then deformably registered to both templates using the `Par0010bspline` parameter file from the elastix Model Zoo, producing two subject-specific transformations and the corresponding displacement fields. Thus, each MCI subject was characterized by a paired deformation profile relative to cognitively normal and dementia references from the same demographic stratum.

3.5 Differential deformation maps

The deformation field was used in this study as an intermediate representation from which regional scalar maps were derived. Let $I_s, T_{g,a}^r : \Omega \subset \mathbb{R}^3 \rightarrow \mathbb{R}$ be the subject image and the selected reference template, respectively, with $r \in \{\text{CN}, \text{DEM}\}$ denoting the cognitively normal and dementia reference conditions. Pairwise deformable registration estimates a transformation $\phi_s^r : \Omega \rightarrow \Omega$ that aligns the moving subject image with the fixed population template. In the notation adopted here, the estimated transformation is written as

$$\phi_s^r(\mathbf{x}) = \mathbf{x} + \mathbf{u}_s^r(\mathbf{x}), \quad (1)$$

where $\mathbf{u}_s^r(\mathbf{x})$ is the displacement vector associated with subject s and reference condition r . The registration problem can be expressed as

$$\hat{\phi}_s^r = \arg \min_{\phi \in \Phi} [\mathcal{D}(T_{g,a}^r, I_s \circ \phi) + \lambda \mathcal{R}_{\text{BS}}(\phi)], \quad (2)$$

where $\mathcal{D}$ is the image dissimilarity term, implemented using Mattes mutual information [30], and $\mathcal{R}_{\text{BS}}$ denotes the smoothness constraint induced by the B-spline transformation model available in ITK-Elastix [17]. In the pairwise deformable registration stage, the `Par0010bspline` parameter file from the elastix Model Zoo was used to estimate the transformations from which the displacement fields were obtained. The symbol $\circ$ denotes function composition; therefore, $(I_s \circ \phi)(\mathbf{x}) = I_s(\phi(\mathbf{x}))$ represents the subject image resampled through the transformation ϕ in the coordinate system of

the fixed template. This formulation was used only to obtain anatomically regularized subject–template correspondences; the full dense vector field was not used directly as the classifier input.

Because the deformation field is estimated from cross-sectional image-to-template registration, the resulting quantities should be interpreted as descriptors of subject–reference anatomical mismatch. They do not represent longitudinal tissue trajectories within the same individual. Moreover, the sign and magnitude of deformation-derived quantities depend on the registration convention and on the selected fixed template. For this reason, the proposed framework uses regional summaries of deformation maps as machine-learning descriptors, rather than interpreting individual voxel values as direct measurements of biological expansion or contraction.

3.5.1 Displacement-amplitude map

The first scalar map extracted from each deformation field was the displacement amplitude:

$$D_s^r(\mathbf{x}) = \|\mathbf{u}_s^r(\mathbf{x})\|_2. \quad (3)$$

This map summarizes the local amount of spatial displacement required to align the subject image to the selected reference template. In the present graph-based framework, the displacement direction was not encoded explicitly. Instead, displacement amplitude was used as a compact regional descriptor of subject–template anatomical deviation, complementary to the Jacobian- and strain-derived maps described below.

3.5.2 Jacobian-based volumetric map

Local volumetric effects induced by the spatial transformation were characterized through the Jacobian determinant. For $\phi_s^r(\mathbf{x}) = \mathbf{x} + \mathbf{u}_s^r(\mathbf{x})$, the Jacobian matrix is

$$\mathbf{J}_{\phi_s^r}(\mathbf{x}) = \nabla \phi_s^r(\mathbf{x}) = \mathbf{I} + \nabla \mathbf{u}_s^r(\mathbf{x}), \quad (4)$$

and the corresponding scalar map is defined as

$$V_s^r(\mathbf{x}) = \det(\mathbf{J}_{\phi_s^r}(\mathbf{x})). \quad (5)$$

The determinant summarizes local volume change associated with the transformation. However, because the transformation is estimated between each subject and a population template, $V_s^r(\mathbf{x})$ is treated here as a deformation-derived volumetric descriptor in template space. It is not interpreted as a direct longitudinal atrophy measure. This distinction is particularly important in the current study, since the same subject is compared with two different reference templates, $T_{g,a}^{\text{CN}}$ and $T_{g,a}^{\text{DEM}}$.

3.5.3 Strain-intensity map

To complement displacement amplitude and local volumetric information, the spatial variability of the displacement field was characterized using the infinitesimal strain

tensor. For each subject and reference template, the tensor was computed as the symmetric component of the displacement-gradient field:

$$\boldsymbol{\varepsilon}_s^r(\mathbf{x}) = \frac{1}{2} \left[\nabla \mathbf{u}_s^r(\mathbf{x}) + (\nabla \mathbf{u}_s^r(\mathbf{x}))^\top \right]. \quad (6)$$

Rather than using the six independent tensor components as separate features, the tensor field was reduced to a scalar strain-intensity map using the Frobenius norm:

$$S_s^r(\mathbf{x}) = \|\boldsymbol{\varepsilon}_s^r(\mathbf{x})\|_F = \sqrt{\sum_{p,q} (\varepsilon_{s,pq}^r(\mathbf{x}))^2}. \quad (7)$$

This scalar map captures the overall magnitude of local stretching and shear encoded in the displacement-gradient field. The use of infinitesimal strain is a pragmatic feature-engineering choice: it provides a low-dimensional descriptor of local deformation intensity, but it should not be interpreted as a complete finite-strain biomechanical model of brain tissue.

3.5.4 ROI-wise aggregation of deformation maps

All deformation-derived scalar maps were represented in the common NAC/template coordinate system, which allowed the same atlas-derived ROI masks to be applied across subjects. For each region-hemisphere pair R_k and each reference condition r , the maps D_s^r , V_s^r , and S_s^r were summarized using four statistics: mean, variance, skewness, and kurtosis. The resulting descriptors capture complementary aspects of subject-template anatomical mismatch: displacement amplitude, local volumetric variation, and deformation-gradient intensity.

This regional aggregation is the methodological link between deformation analysis and the graph-based classifier. The dense deformation fields are not treated as images to be classified directly. Instead, they are converted into anatomically organized node attributes, where each graph node corresponds to a specific ROI and hemisphere. This design differentiates the present work from descriptor-based brain-age modeling: the deformation field is used here to generate regional attributes for inter-regional graph learning in the sMCI versus pMCI classification task.

3.6 Feature extraction

Feature extraction was performed after the deformation fields had been converted into the three scalar maps defined in Subsection 3.5. Bilateral analyses were conducted over 15 predefined anatomical regions, with left and right hemispheres treated independently. For each region-hemisphere pair, three deformation-derived maps were considered: displacement amplitude, Jacobian-based volumetric variation, and strain intensity.

From each map, four statistical descriptors—mean, variance, skewness, and kurtosis—were computed inside the corresponding ROI. The same procedure was repeated for the two matched reference conditions, cognitively normal and dementia.

Therefore, the feature space was defined by 15 regions, two hemispheres, three deformation maps, four regional statistics, and two reference templates, resulting in 720 region-based attributes per subject. These attributes were subsequently organized as node features for the graph-based classification framework.

3.7 Feature selection

Feature selection was used to reduce redundancy and to evaluate whether a compact subset of deformation-derived attributes improved classification performance. A correlation-based filtering step was first applied to remove highly redundant attributes, using a Pearson correlation threshold of 0.95. This conservative threshold was chosen to eliminate near-duplicated information while preserving the diversity of the deformation descriptors. After this step, the feature representation was reduced from 720 attributes to 656 attributes for the male cohort and 654 attributes for the female cohort.

The effect of additional feature selection was then investigated using the **SelectKBest** univariate feature-selection procedure with mutual information as the scoring function, as implemented in scikit-learn [28]. Different numbers of top-ranked features were evaluated to characterize the relationship between feature-set size and AUC performance, as shown in Fig. 3. The best performance was obtained when using the full set of features retained after correlation filtering. This result indicates that the deformation-derived representation contains complementary information distributed across several descriptors and regions, rather than being concentrated in a small subset of attributes.

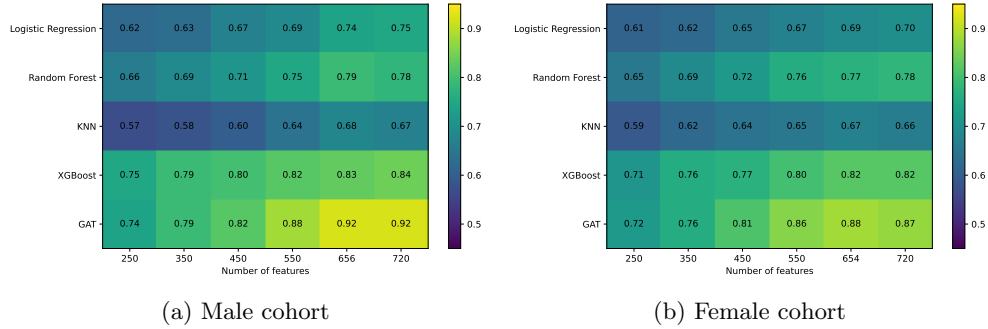


Fig. 3: Heatmaps illustrating the impact of the number of selected features on the AUC performance across different models for (a) male and (b) female cohorts.

Therefore, the final representation retained the correlation-filtered feature set. This choice reduced redundancy and potential multicollinearity while preserving discriminative information. Feature filtering and model selection were incorporated into the nested cross-validation protocol described in Subsection 3.10.

3.8 Graph construction

Each subject was represented as a graph $G = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the set of nodes and $\mathcal{E}$ is the set of edges. Nodes correspond to atlas-defined brain regions separated by hemisphere. The selected ROIs generated 30 nodes per subject. For each node, the associated feature vector contained the region-wise statistical descriptors extracted from the displacement-amplitude, Jacobian-based volumetric, and strain-intensity maps, computed with respect to the cognitively normal and dementia reference templates. Thus, each subject was described by a node-feature matrix $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times d}$ containing subject-specific deformation information.

A fixed population-level graph template was used to provide a common anatomical domain for message passing across all subjects. The template was constructed from inter-regional correlation patterns in a label-free manner: class labels, subject-level predictions, and classification performance metrics were not used to define candidate edges. The graph therefore encodes recurrent associations among regional deformation descriptors, while the subject-specific node-feature matrix carries the information used by the classifier for sMCI versus pMCI discrimination.

For graph estimation, node-level representations were first organized across subjects within each graph-estimation fold. Pairwise Pearson correlation coefficients r_{ij} were then computed between node representations to quantify the similarity of deformation patterns between regions i and j . Candidate undirected edges were defined using an explicit absolute-correlation criterion: an edge between nodes i and j was considered present in a given graph-estimation fold when $|r_{ij}| \geq 0.15$. This threshold served as an initial filter for very weak associations and did not impose a top- k rule, a statistical-significance test, or a fixed edge density at the fold level.

To retain recurrent inter-regional associations, graph estimation used a 10-fold stability procedure. Let $K = 10$ be the number of graph-estimation folds. The stability frequency of each edge was computed as

$$f_{ij} = \frac{1}{K} \sum_{k=1}^K \mathbb{I}(|r_{ij}^{(k)}| \geq 0.15), \quad (8)$$

where $\mathbb{I}(\cdot)$ is the indicator function.

The final graph retained only edges satisfying $f_{ij} \geq \tau$, where $\tau \in \{0.50, 0.60, 0.70, 0.80, 0.90\}$ was evaluated to assess the effect of graph sparsity. Therefore, the 60% graph template corresponds to edges present in at least 6 of the 10 graph-estimation folds; it does not denote the top 60% most frequent edges.

Separate graph templates were estimated for male and female cohorts using the same procedure. This design allows the graph structure to reflect cohort-specific inter-regional deformation patterns while keeping the node definitions, feature construction, and classification architecture identical across groups. Retained edges were treated as undirected, and edge attributes were represented by the corresponding Pearson correlation values. Self-loops were added to preserve node-specific information during message passing. The resulting fixed graph topology, combined with subject-specific node attributes, served as input to the GAT classifier.

Fig. 4 illustrates the fixed graph template from complementary axial, sagittal, and coronal views, making explicit the anatomical domain on which message passing is performed. In this representation, nodes correspond to the selected bilateral ROIs and edges correspond to stable inter-regional deformation associations retained after the graph-estimation procedure; therefore, the figure visually links the correlation-based topology construction to the subsequent GAT-based integration of distributed structural information.

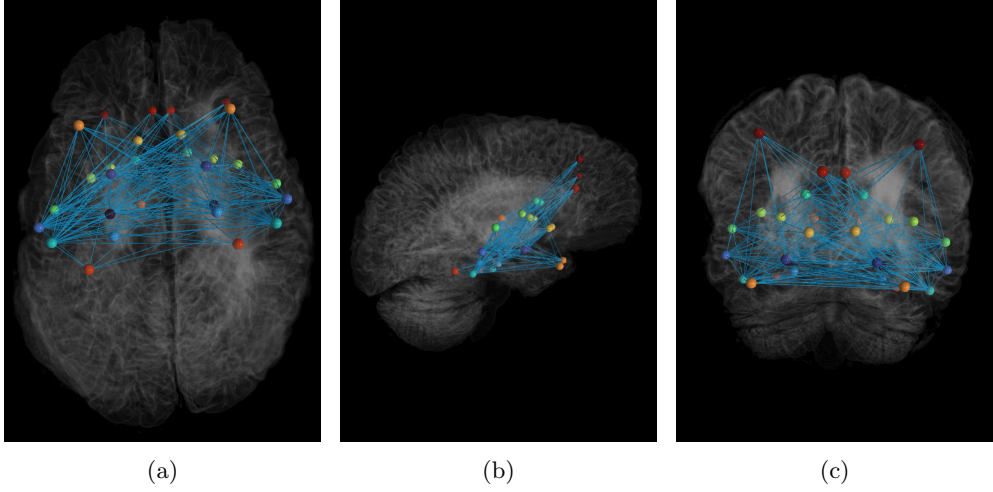


Fig. 4: Visualization of the fixed graph template across anatomical views: (a) axial, (b) sagittal, and (c) coronal.

3.9 Graph Attention-Based Classification Model

To model inter-regional relationships encoded in the fixed graph template, we employed a graph-based deep learning architecture based on graph message passing and attention mechanisms. Let $G = (\mathcal{V}, \mathcal{E})$ denote the fixed graph topology shared across all subjects, where $|\mathcal{V}|$ corresponds to the number of brain regions and $\mathcal{E}$ represents the set of stable edges identified during graph construction. For each subject, node attributes are represented by a feature matrix $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times d}$, while the graph structure $\mathcal{E}$ remains unchanged.

The proposed architecture leverages the GAT mechanism [33] to adaptively weight the influence of neighboring nodes during message passing. Given a node i and its neighborhood $\mathcal{N}(i)$, the attention coefficient between node i and a neighboring node j is computed as

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_i \parallel \mathbf{W}\mathbf{h}_j \parallel e_{ij}]))}{\sum_{k \in \mathcal{N}(i)} \exp(\text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_i \parallel \mathbf{W}\mathbf{h}_k \parallel e_{ik}]))}, \quad (9)$$

where $\mathbf{h}_i$ denotes the feature vector of node i , $\mathbf{W}$ is a learnable linear transformation, $\mathbf{a}$ is the attention vector, $\parallel$ denotes vector concatenation, and e_{ij} represents the scalar edge attribute associated with the connection between nodes i and j .

The updated node representation is obtained by aggregating the transformed features of neighboring nodes weighted by the learned attention coefficients:

$$\mathbf{h}'_i = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} \mathbf{W} \mathbf{h}_j \right), \quad (10)$$

where $\sigma(\cdot)$ denotes a nonlinear activation function. Multi-head attention is employed in the first graph attention layer to improve model expressiveness and stabilize training.

The network consists of two graph attention layers followed by batch normalization and dropout regularization. To obtain a graph-level representation suitable for subject classification, a global mean pooling operation is applied over all node embeddings as

$$\mathbf{z} = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} \mathbf{h}'_i, \quad (11)$$

and the pooled embedding $\mathbf{z}$ is then passed to a fully connected layer to predict the subject-level class label.

Importantly, while node features are subject-specific, the graph connectivity structure is fixed and identical for all subjects, as described in Section 3.8. This design ensures that classification performance is driven by differences in regional deformation patterns rather than variations in graph topology. The GAT model is trained end-to-end using cross-entropy loss, and optimization is performed using the Adam optimizer [15].

3.10 Classification model and nested validation protocol

In this work, we compared the performance of Logistic Regression, Random Forest, K-Nearest Neighbors (KNN), XGBoost [6], and the proposed GAT-based classifier. Model development and performance estimation were organized using a nested cross-validation strategy [5], illustrated in Fig. 5. This design separates model selection from final performance estimation and provides a controlled protocol for comparing the GAT with conventional classifiers trained on the same deformation-derived regional descriptors.

The outer loop consisted of a 10-fold cross-validation procedure. In each outer iteration, one fold was held out for testing, whereas the remaining nine folds formed the corresponding outer training set. The inner loop consisted of 5-fold cross-validation performed within each outer training set. Correlation filtering, **SelectKBest** feature selection, feature standardization when applicable, and hyperparameter tuning were fitted and evaluated inside this model-selection loop. Inner-validation results were used only to select the configuration adopted in the corresponding outer iteration.

After model selection, the selected preprocessing, feature-selection, and classifier configuration was refitted using the complete outer training set and then evaluated

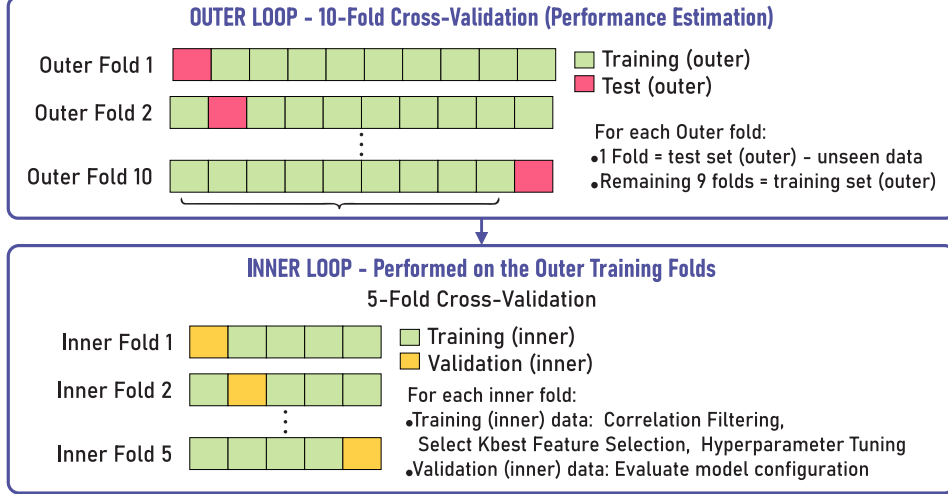


Fig. 5: Nested cross-validation scheme adopted in the proposed framework. The outer 10-fold loop was used for final performance estimation, with one fold held out as test data in each iteration. The inner 5-fold loop was performed within the corresponding outer training folds and was used for correlation filtering, **SelectKBest** feature selection, and hyperparameter tuning.

once on the held-out outer test fold. This procedure produced one test estimate per outer fold, resulting in ten outer-fold estimates for each reported metric. Graph construction followed the label-free stability procedure described in Subsection 3.8.

The GAT-based classifier was implemented using PyTorch [27] and PyTorch Geometric [9]. The architecture employed two GATv2 convolutional layers [3]. The selected configuration obtained during the inner-loop hyperparameter optimization employed three attention heads in the first layer with a hidden dimension of 32, followed by batch normalization, ReLU activation, and dropout. The second attention layer used a single attention head and preserved a 32-dimensional hidden representation before global mean pooling and final classification. A dropout rate of 0.3 was used throughout the network.

Model optimization was performed using the Adam optimizer. The selected hyperparameter configuration employed a learning rate of 8×10^{-4} and weight decay of 5×10^{-3} . Models were trained with a batch size of 32 for a maximum of 200 epochs. Early stopping was applied after a minimum of 20 epochs using a patience of 5 validation checks and a minimum improvement threshold of 5×10^{-4} . The monitored validation score was defined as $0.7 \times \text{AUC} + 0.3 \times \text{AP}$, where AP denotes average precision. During training, DropEdge [31] with probability 0.3 was used as an additional regularization strategy. Random seeds were fixed for data splitting and model training to improve reproducibility.

All models were optimized through Grid Search within the inner loop. The evaluated hyperparameter search spaces for the GAT architecture and baseline models are summarized in Table 2.

Table 2: Hyperparameter search space evaluated during model selection.

Model	Hyperparameter	Evaluated values
GAT	hidden dimension	{16, 32, 64}
	attention heads	{1, 2, 3, 4}
	learning rate	$\{1 \times 10^{-3}, 8 \times 10^{-4}, 5 \times 10^{-4}\}$
	weight decay	$\{1 \times 10^{-4}, 1 \times 10^{-3}, 5 \times 10^{-3}\}$
	batch size	{16, 32}
LR	penalty	{l2}
	C	{0.01, 0.1, 1, 10, 100}
Random Forest	n-estimators	{100, 200, 500, 1000}
	max-depth	{None, 5, 10, 30, 50, 70, 100}
	min-samples-split	{2, 5, 10, 20, 40, 50}
	min-samples-leaf	{1, 2, 4, 6, 8, 10}
KNN	n-neighbors	{3, 5, 7, 9, 11}
	weights	{uniform, distance}
	metric	{euclidean, manhattan}
XGBoost	n-estimators	{100, 200, 500, 700, 1000}
	max-depth	{3, 5, 7, 10, 30, 50, 70, 100}
	learning rate	{0.01, 0.05, 0.1}
	subsample	{0.5, 0.6, 0.7, 0.8, 1.0}

Feature standardization using z-score normalization (StandardScaler) was applied within the appropriate training partitions for scale-sensitive classifiers, including Logistic Regression and KNN. The same transformation parameters estimated from the corresponding training data were subsequently applied to validation or test partitions, preserving the separation between training and evaluation stages.

No explicit class balancing strategy was applied, as the class distribution remained relatively balanced across the analyzed cohorts. Final performance metrics were obtained by averaging the results across the ten outer test folds. Among the evaluated methods, the GAT-based classifier achieved the best overall results, yielding the highest AUC and classification performance. Consequently, this model was selected as the primary classifier for subsequent analyses.

4 Results and Discussions

This section presents the experimental results for the sMCI versus pMCI classification task, emphasizing the influence of graph sparsity, the comparison with conventional machine learning baselines trained on the same deformation-derived regional attributes, and the contribution of the proposed graph-attentional representation to modeling inter-regional structural relationships.

4.1 sMCI vs. pMCI Classification

The proposed framework was evaluated on the task of discriminating sMCI from pMCI using sex-specific fixed graph templates and region-wise deformation features. This classification setting is demanding because progression-related structural differences are expected to be subtle, heterogeneous, and distributed across multiple anatomical

regions. The results were therefore analyzed from two complementary perspectives: the influence of graph sparsity on GAT performance and the comparison between the graph-based representation and conventional classifiers trained on the same regional descriptors.

Performance was first analyzed as a function of the edge-frequency threshold used during fixed template construction, as shown in Tables 3 and 4. For both male and female cohorts, intermediate graph densities produced the highest observed performance, with the best results obtained when retaining edges appearing in at least 60% of the graph-estimation folds. This behavior indicates that preserving recurrent inter-regional associations is beneficial, whereas overly sparse or overly dense templates may reduce the effectiveness of message passing.

Using the 60% frequency template, the proposed GAT-based classifier achieved an AUC of 0.92 ± 0.03 , an accuracy of $81.21 \pm 4.47\%$, and an F1-score of 0.79 ± 0.04 in the male cohort. In the female cohort, the same configuration yielded an AUC of 0.88 ± 0.03 , an accuracy of $78.41 \pm 4.39\%$, and an F1-score of 0.73 ± 0.07 . The corresponding confidence intervals are reported in Tables 3 and 4.

To contextualize these findings, the proposed model was compared with conventional machine learning classifiers trained on the same region-wise features but without explicit graph message passing, as summarized in Tables 5 and 6. Logistic Regression, K-Nearest Neighbors, Random Forest, and XGBoost served as baseline methods. Among these, XGBoost achieved the strongest baseline performance; however, the GAT achieved the highest mean AUC, accuracy, and F1-score in both cohorts. The corresponding receiver operating characteristic (ROC) curves are presented in Fig. 6. McNemar’s test was applied to subject-level hard predictions aggregated across the external folds of the nested cross-validation procedure and indicated statistically significant differences between the GAT and the evaluated baseline models ($p < 0.01$).

Table 3: Classification results for male subjects using different edge frequency thresholds (mean $\pm$ std and 95% CI).

Edge Freq. (%)	AUC		Accuracy (%)		F1-Score	
	Mean $\pm$ SD	95% CI	Mean $\pm$ SD	95% CI	Mean $\pm$ SD	95% CI
50	0.89 $\pm$ 0.03	0.87–0.91	77.86 $\pm$ 5.01	74.28–81.44	0.75 $\pm$ 0.04	0.72–0.78
60	0.92 $\pm$ 0.03	0.90–0.94	81.21 $\pm$ 4.47	78.01–84.41	0.79 $\pm$ 0.04	0.76–0.82
70	0.86 $\pm$ 0.04	0.83–0.89	73.96 $\pm$ 4.18	70.97–76.95	0.74 $\pm$ 0.05	0.70–0.78
80	0.87 $\pm$ 0.03	0.85–0.89	75.28 $\pm$ 4.63	71.97–78.59	0.65 $\pm$ 0.05	0.61–0.69
90	0.85 $\pm$ 0.03	0.83–0.87	64.79 $\pm$ 5.28	61.01–68.57	0.61 $\pm$ 0.05	0.57–0.65

Table 4: Classification results for female subjects using different edge frequency thresholds (mean $\pm$ std and 95% CI).

Edge Freq. (%)	AUC		Accuracy (%)		F1-Score	
	Mean $\pm$ SD	95% CI	Mean $\pm$ SD	95% CI	Mean $\pm$ SD	95% CI
50	0.86 $\pm$ 0.03	0.84–0.88	74.29 $\pm$ 6.52	69.63–78.95	0.71 $\pm$ 0.06	0.67–0.75
60	0.88 $\pm$ 0.03	0.86–0.90	78.41 $\pm$ 4.39	75.27–81.55	0.73 $\pm$ 0.07	0.68–0.78
70	0.85 $\pm$ 0.04	0.82–0.88	70.82 $\pm$ 6.39	66.25–75.39	0.69 $\pm$ 0.06	0.65–0.73
80	0.84 $\pm$ 0.06	0.80–0.88	65.82 $\pm$ 6.19	61.39–70.25	0.63 $\pm$ 0.07	0.58–0.68
90	0.84 $\pm$ 0.04	0.81–0.87	62.81 $\pm$ 5.38	58.96–66.66	0.60 $\pm$ 0.04	0.57–0.63

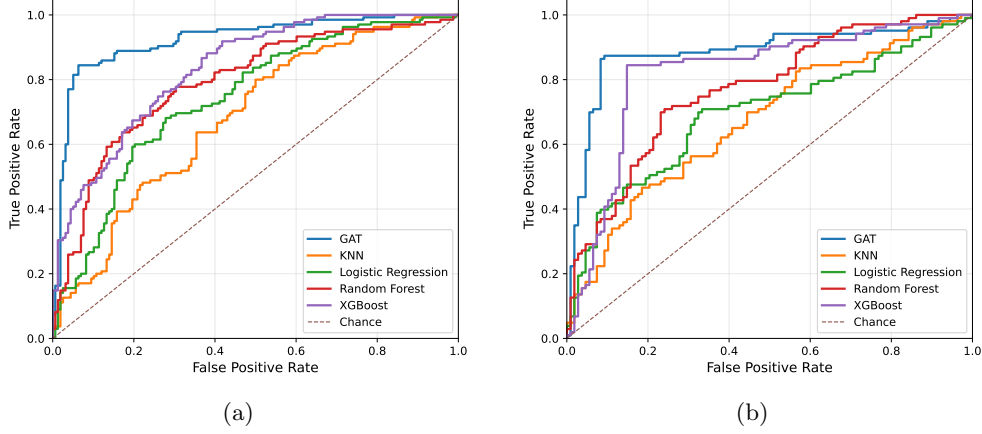


Fig. 6: ROC curves for sMCI vs. pMCI classification using the proposed framework. (a) Male cohort. (b) Female cohort.

Table 5: Performance comparison for male subjects (mean $\pm$ std and 95% CI).

Model	AUC		Accuracy (%)		F1-Score	
	Mean $\pm$ SD	95% CI	Mean $\pm$ SD	95% CI	Mean $\pm$ SD	95% CI
Logistic Regression	0.74 $\pm$ 0.05	0.70–0.78	62.39 $\pm$ 6.42	57.80–66.98	0.58 $\pm$ 0.06	0.54–0.62
Random Forest	0.79 $\pm$ 0.04	0.76–0.82	74.44 $\pm$ 6.09	70.08–78.80	0.69 $\pm$ 0.07	0.64–0.74
KNN	0.68 $\pm$ 0.05	0.64–0.72	63.21 $\pm$ 5.26	59.45–66.97	0.64 $\pm$ 0.03	0.62–0.66
XGBoost	0.83 $\pm$ 0.05	0.79–0.87	75.25 $\pm$ 5.46	71.34–79.16	0.72 $\pm$ 0.06	0.68–0.76
GAT	0.92 $\pm$ 0.03	0.90–0.94	81.21 $\pm$ 4.47	78.01–84.41	0.79 $\pm$ 0.04	0.76–0.82

Table 6: Performance comparison for female subjects (mean $\pm$ std and 95% CI).

Model	AUC		Accuracy (%)		F1-Score	
	Mean $\pm$ SD	95% CI	Mean $\pm$ SD	95% CI	Mean $\pm$ SD	95% CI
Logistic Regression	0.69 $\pm$ 0.03	0.67–0.71	59.83 $\pm$ 4.62	56.53–63.13	0.59 $\pm$ 0.04	0.56–0.62
Random Forest	0.77 $\pm$ 0.04	0.74–0.80	72.35 $\pm$ 4.36	69.23–75.47	0.67 $\pm$ 0.05	0.63–0.71
KNN	0.67 $\pm$ 0.05	0.63–0.71	64.62 $\pm$ 3.81	61.89–67.35	0.63 $\pm$ 0.04	0.60–0.66
XGBoost	0.82 $\pm$ 0.03	0.80–0.84	73.71 $\pm$ 4.76	70.30–77.12	0.67 $\pm$ 0.05	0.63–0.71
GAT	0.88 $\pm$ 0.03	0.86–0.90	78.41 $\pm$ 4.39	75.27–81.55	0.73 $\pm$ 0.07	0.68–0.78

The performance advantage of the GAT over vector-based classifiers suggests that discriminative information is not restricted to isolated regional summaries. Instead, the organization of deformation descriptors across connected anatomical regions contributes to the classification process. This observation is consistent with the motivation of the proposed framework: to move from independent ROI measurements toward a structured representation in which regional deformation patterns and their inter-regional relationships are jointly modeled.

The sensitivity analysis over edge-frequency thresholds further supports the relevance of graph organization. The 60% graph template provided the best empirical balance between sparsity and information flow, whereas more restrictive or more permissive templates produced lower mean performance. Together with the comparison against conventional classifiers trained on the same features, these results indicate

that the graph-attentional representation provides added value beyond the vectorized feature representation.

To further contextualize the proposed approach with respect to the state of the art, Table 7 summarizes recent studies on sMCI vs. pMCI classification using structural MRI and multimodal data.

As shown, existing methods report strong performance, particularly those based on deep learning, attention mechanisms, and multimodal fusion. For instance, Luo et al. [19] and Wei et al. [34] report competitive AUC and accuracy values using attention-based architectures, while Hu et al. [12] and Xu et al. [36] explore hybrid CNN and attention mechanisms. The present framework achieves competitive results using T1-weighted MRI descriptors organized through an explicit graph representation of inter-regional deformation relationships.

Because the studies listed in Table 7 differ in sample size, age range, input modalities, clinical variables, longitudinal requirements, and validation protocols, the table should be read as contextual information rather than a controlled head-to-head benchmark. The controlled comparison in the present work is the internal comparison with baseline classifiers trained on the same deformation-derived regional attributes and evaluated under the same nested cross-validation protocol.

An additional benefit of the proposed framework is its anatomically organized representation. Graph nodes correspond to predefined brain regions, and edges encode stable inter-regional associations derived from deformation features. This representation enables post hoc inspection of graph organization and regional interactions while maintaining the primary focus on predictive classification.

It is important to note that, while most related studies consider an age range of 55–90 years, the present work restricts the analysis to subjects aged 60–90 years. This choice was motivated by the requirements of the template construction stage. Specifically, the OASIS dataset used for template generation contains a limited number of dementia cases in the 55–60 age interval, which is insufficient to support reliable population templates, particularly under stratification by sex and diagnostic group. Therefore, the adopted age range ensures more robust and consistent deformation-based analysis.

4.2 Comparative Sex-Based Trends

Building upon the fixed graph structures described in Section 3.8, we conducted a comparative analysis of sex-specific inter-regional organization patterns observed during graph estimation. The graph construction procedure, stability thresholds, and region definitions were identical across male and female cohorts. Consequently, differences between the resulting graph templates reflect cohort-specific correlation patterns estimated under the same methodological conditions.

Across both cohorts, classification performance depended on the edge-frequency threshold, with the highest observed performance obtained using the 60% template. Male subjects reached an AUC of 0.92 ± 0.03 and an accuracy of $81.21 \pm 4.47\%$, whereas female subjects achieved an AUC of 0.88 ± 0.03 and an accuracy of $78.41 \pm 4.39\%$. Since node features, model architecture, and training protocol were kept consistent, these

Table 7: Comparison of pMCI vs. sMCI classification methods: performance across models, modalities, and datasets.

References	Images	Age	Model	Modality	Database	Results
[24]	264	[55–90]	LDS + Random Forest	T1-w	ADNI	AUC: 0.90, Accuracy: 82.00%
[20]	264	[55–90]	Multi-stream CNN (patch-based + transfer learning)	T1-w	ADNI	AUC: 0.87
[14]	684	[55–90]	DenseNet + BiLSTM (longitudinal)	T1-w	ADNI	ACC: 83.51%
[19]	593	[55–90]	Dual-attention fusion network (CNN + Transformer)	T1-w + clinical	ADNI	AUC: 0.874, Accuracy: 81.34%
[37]	921	[55–90]	Transformer-based multimodal model (AD-Transformer)	MRI + clinical + genetic data	ADNI	AUC: 0.845, Accuracy: 75.30%
[12]	823	[55–90]	MMS-TransEtnet (CNN + Transformer)	T1-w	ADNI	AUC: 0.86, Accuracy: 87.28%
[36]	517	[55–90]	Multi-view CNN + attention fusion	T1-w	ADNI	AUC: 0.88, Accuracy: 84.6%
[7]	526	[55–90]	Multi-stage ROI-based fusion CNN (m2ROI-FN)	T1-w + PET	ADNI	Accuracy: 77.40%
[34]	788	[55–90]	3D decoupling self-attention network	T1-w	ADNI / AIBL	AUC: 0.88, Accuracy: 89.40%
Our method	469	[60–90]	Graph-based multi-region framework (deformation fields + Jacobian + strain + GAT)	T1-w	ADNI (classification); OASIS (templates)	Male / Female AUC: 0.92 / 0.88 ACC: 81.21% / 78.41% F1: 0.79 / 0.73

results indicate that sex-specific graph templates provide a useful way to organize cohort-dependent deformation relationships.

The graph templates also exhibited distinct organizational patterns. In the male graph template, high-degree nodes were concentrated in a limited subset of regions, with the *right inferior temporal gyrus* emerging as the most connected node, followed by the *left middle temporal gyrus* and the *right amygdala*. This configuration reflects a more selective inter-regional organization pattern, characterized by fewer dominant hubs and greater variability in node degree.

In contrast, the female graph template exhibited a more homogeneous degree distribution, with multiple regions reaching the maximum observed degree. Highly connected nodes included the *right inferior temporal gyrus*, *right middle temporal gyrus*, *bilateral parahippocampal gyri*, and the *left inferior temporal gyrus*. This pattern indicates a denser and more uniformly distributed inter-regional organization in the estimated graph template for the female cohort.

These sex-specific trends reinforce the value of population-aware graph construction. Rather than relying on a single graph for all subjects, the proposed framework allows inter-regional deformation relationships to be organized according to the cohort being analyzed, providing a flexible representation for studying structural patterns associated with MCI progression.

4.3 Statistical analysis

Model performance was estimated from the outer loop of the nested cross-validation procedure. The ten outer test folds provided the only scores used to compute the final reported performance. The five inner folds were used exclusively for model selection and hyperparameter tuning and were not treated as independent test estimates. Results are therefore reported as mean $\pm$ standard deviation across the ten outer test folds.

To assess the reliability of these estimates, 95% confidence intervals were computed using the Student’s t-distribution over the outer-fold scores. Confidence intervals were calculated as:

$$\bar{x} \pm t_{(0.975, n-1)} \cdot \frac{s}{\sqrt{n}}, \quad (12)$$

where $\bar{x}$ is the mean performance across the outer test folds, s is the corresponding standard deviation, and $n = 10$ is the number of outer folds. All analyses were conducted separately for male and female cohorts. These intervals summarize variability across the outer test folds and provide a descriptive measure of cross-validation stability.

4.4 Interpretability and Clinical Relevance

The proposed framework is primarily a classification model, but its graph-based formulation provides an anatomically organized and inspectable representation of the input data. Each node corresponds to a predefined brain region, and the fixed graph template defines explicit inter-regional relationships estimated from deformation-derived features. This structure is more transparent than a purely unstructured feature

vector because the model operates over anatomically meaningful units and stable inter-regional associations.

The attention mechanism of the GAT adaptively weights neighboring regions during message passing. These coefficients can be used for exploratory post hoc inspection of information flow through the graph. In this sense, attention complements the anatomical organization of the model by indicating which neighboring regions receive greater emphasis during graph-based aggregation.

The analysis of node degree distributions indicated that medial temporal and adjacent temporal structures were frequently involved in highly connected regions across the estimated graph templates. These included the hippocampus, amygdala, inferior temporal gyrus, middle temporal gyrus, parahippocampal gyrus, fusiform gyrus, and temporal pole. The involvement of these structures is consistent with the known relevance of medial temporal and temporo-limbic regions in AD-related neurodegeneration, supporting the anatomical plausibility of the graph organization observed in the experiments.

From a translational perspective, the main value of the framework lies in combining predictive modeling with an explicit anatomical representation. The model does not reduce the brain to an unordered feature vector; instead, it organizes deformation-derived regional descriptors in a graph where local morphometric information and inter-regional relationships can be analyzed jointly.

4.5 Methodological considerations and future work

The experimental design was organized to provide a controlled assessment of the proposed framework under a nested cross-validation protocol. In this setting, the outer loop was used for performance estimation, whereas model selection and hyperparameter tuning were performed within the inner loop. This organization supports a consistent comparison between the GAT and conventional machine learning baselines trained on the same deformation-derived regional descriptors.

A key methodological aspect of the study is the use of fixed, stability-based graph templates. The graph was constructed in a label-free manner and was designed to encode recurrent inter-regional associations among anatomical regions. Thus, the graph defines the message-passing domain of the GAT, while subject-specific classification is driven by the nodal feature matrices derived from each MRI scan. This formulation preserves the anatomical organization of regional deformation descriptors and allows the model to integrate information across connected regions.

The evaluation of different edge-frequency thresholds characterized the effect of graph sparsity on classification performance. Among the analyzed configurations, retaining edges observed in at least 60% of the graph-estimation folds provided the best empirical balance between sparsity and information flow. This result supports the use of a stability criterion for retaining recurrent anatomical associations rather than relying on a fully connected graph or on a single correlation estimate.

The framework is also extensible. Future work can evaluate the same representation in independent cohorts, incorporate additional imaging modalities or longitudinal measurements, and further explore the biological interpretation of graph organization

patterns. These extensions are natural next steps for a methodology that already combines deformation-based structural descriptors, population-level graph templates, and graph attention learning within a single MRI-based classification pipeline.

5 Conclusions

This work introduced a graph-based multi-region framework for MRI-based classification of pMCI using deformation-derived structural descriptors. The proposed approach integrates displacement-amplitude, Jacobian-based volumetric, and strain-intensity information extracted from T1-weighted MRI into anatomically organized graph representations. By combining these regional descriptors with GAT-based learning, the method explicitly models inter-regional deformation relationships rather than treating brain regions as independent or unstructured features.

The proposed graph-attentional representation achieved the highest mean performance among the evaluated models in both male and female cohorts. The best configuration retained edges present in at least 60% of the graph-estimation folds, indicating that a stability-based graph construction strategy can provide an effective balance between anatomical organization, sparsity, and discriminative information flow. The comparison with Logistic Regression, KNN, Random Forest, and XGBoost trained on the same regional attributes further supports the benefit of incorporating graph structure into the representation.

A distinctive contribution of the study is the integration of deformation-based MRI biomarkers with fixed population-level graph templates. This formulation provides an anatomically meaningful bridge between local morphometric descriptors and distributed inter-regional patterns, allowing the classifier to exploit both regional structural alterations and their relationships across medial temporal and temporo-limbic areas. The use of sex-specific graph templates also revealed different organizational patterns between male and female cohorts, demonstrating the flexibility of the framework for population-aware structural analysis.

Overall, the proposed framework offers a principled strategy for organizing deformation-derived MRI information in sMCI versus pMCI classification. Its main contribution lies in moving from isolated regional measurements to a structured representation in which stable inter-regional associations guide graph-based learning. These findings support deformation-informed anatomical graphs as a promising direction for computational neuroimaging studies of MCI progression.

Declarations.

Author contributions. **Leandro Prado de Andrade:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Mario Augusto de Souza Liziér:** Conceptualization, Visualization, Methodology, Writing – review & editing. **Ricardo José Ferrari:** Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Writing – original draft, Writing – review & editing.

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Data availability. The data analyzed in this study are available from the Alzheimer’s Disease Neuroimaging Initiative (ADNI) database (<https://adni.loni.usc.edu/>) and from the Open Access Series of Imaging Studies (OASIS) project (<https://sites.wustl.edu/oasisbrains/>), subject to their respective data-use and access procedures.

Code availability. The code used for the analyses is available from the corresponding author upon reasonable request.

Competing interests. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethics approval. This study is a secondary analysis of de-identified data obtained from ADNI and OASIS. No new human participants, biological samples, or animal experiments were collected or performed by the authors for this manuscript. Ethical approval procedures for the original data collection are governed by the respective ADNI and OASIS study protocols and data-use agreements.

Consent to participate. Not applicable to this secondary analysis by the authors; informed-consent procedures for the original data collection are governed by the respective ADNI and OASIS study protocols and data-use agreements.

Consent for publication. Not applicable.

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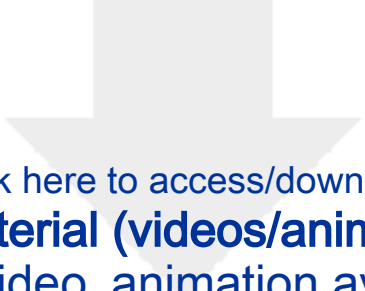
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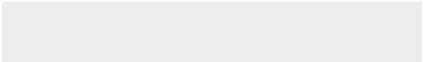
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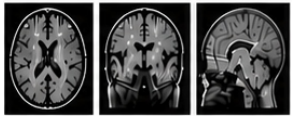
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Supplementary material (videos/animations/long tables)
video_animation.avi



MRI PREPROCESSING

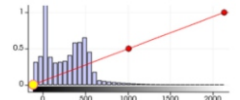
MRI T1-weighted Images



Preprocessing

- Skull Stripping
- Denoising
- N4 Bias Field Correction

Intensity Normalization



Spatial Normalization

- Affine registration to NAC Template

Population-Specific Templates

- Sex-Specific groupwise template construction

DEFORMATION FEATURE EXTRACTION



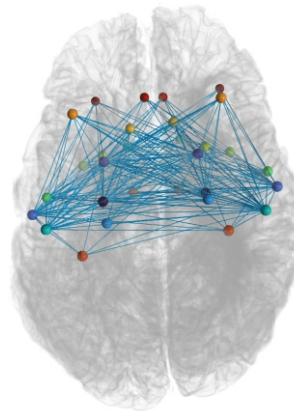
ROI Feature Extraction

- Accumbens
- Amygdala
- Anterior cingulate gyrus
- Caudate
- Fusiform gyrus
- Hippocampus
- Inferior temporal gyrus
- Insula
- Middle frontal gyrus
- Middle temporal gyrus
- Parahippocampal gyrus
- Putamen
- Superior temporal gyrus
- Temporal pole
- Thalamus

GRAPH CONSTRUCTION

Graph Definition

- Pearson Correlation between ROIs
- Edge Stability Thresholding (60%)
- Fixed Graph Topology



FEATURE / MODEL SELECTION & EVALUATION

Feature Selection

- Correlation Filtering
- SelectKBest (Mutual Information)

Model Evaluation

- Nested Cross-Validation
 - Outer Loop: 10-fold
 - Inner Loop :5-fold
- Hyperparameter Tuning
- Model Assessment

CLASSIFICATION

Graph Neural Network

- Graph Attention Network (GAT)

Performance Evaluation

- ROC Curve & AUC
- Accuracy & F1-score

Final Prediction

sMCI or pMCI